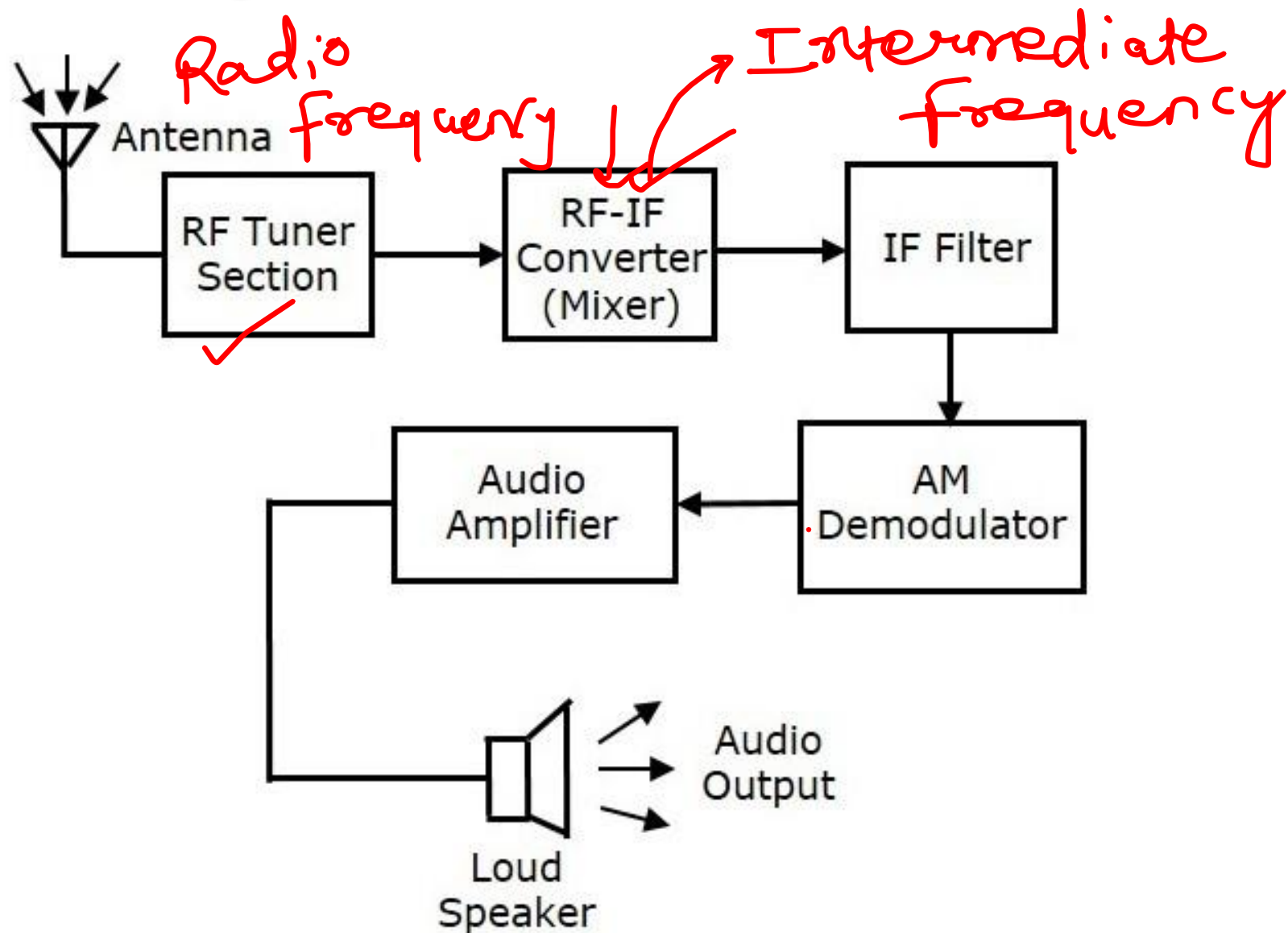


Chapter 4

Radio Receivers

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Block diagram of General receiver



Concept of Intermediate Frequency (IF)

- **The Master- Chef Analogy**

- Imagine a Master Chef who needs to cook 100 different types of exotic vegetables (the Radio Signals)

- **The Problem:** Every vegetable has a different shape, skin thickness, and size. If the chef had to build custom peeling machine for every single vegetable, the kitchen would be massive and expensive.

Concept of Intermediate Frequency (IF)

- **The IF Solution:**

- Instead, the chef hires an assistant to “pre-process” every vegetable into standard 1-inch cube (**the intermediate frequency**)

- **The Benefit:** Now, the chef only needs one high-quality machine that processes 1-inch cubes. It doesn't matter if it started as pumpkin or a carrot; by the time it reaches the chef, it's a standard size.

Concept of Intermediate Frequency (IF)

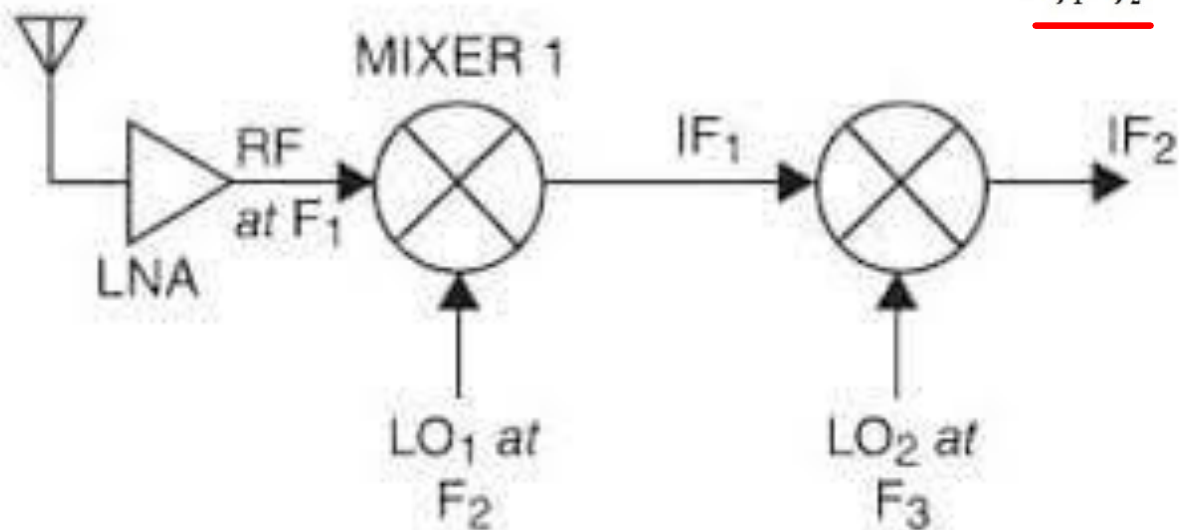
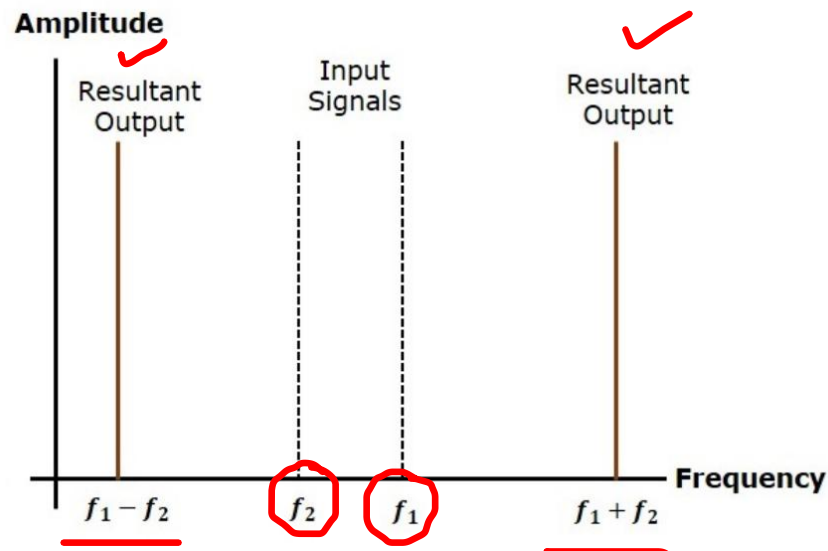
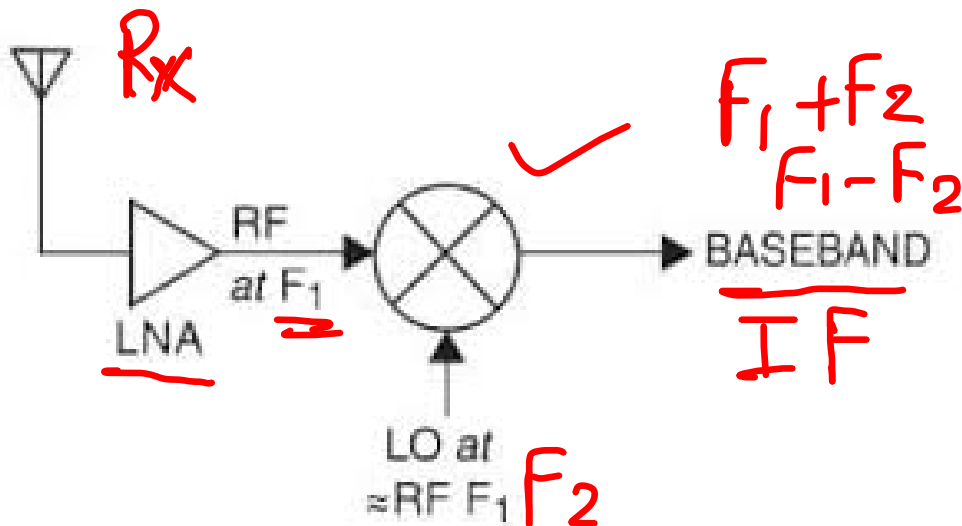
- **In Radio Terms:**

- We convert high-frequency signals (88 MHz to 108 MHz) down to a Fixed Standard Frequency (10.7 MHz for FM)
- So our filters and amplifiers can be optimized for just that one number

AM and FM Ranges

- AM Range: 540 to 1600 KHz
- AM IF: 455 KHz ✓
- FM Range: 88 to 108 MHz
- FM IF: 10.7 MHz ✓

Mixer/Converter Section



Mixer/Converter Section

- The receiver is tuned to 550KHz. *Incoming RF*

- The LO will generate 1005 KHz. (455KHz above station carrier frequency).

(Heterodyning)

- The mixer will produce 455KHz. (1005-550).

- The output of IF amplifier will respond to 455KHz $\pm$ 3KHz. Other frequencies rejected.

- The converted signal is rectified, filtered (detected), and amplified for listening purpose.

Receiver Parameters / Characteristic

- Selectivity
- Fidelity
- Image frequency and its rejection
- Bandwidth Improvement
- Sensitivity
- Dynamic Range
- Insertion Loss

Selectivity

- It is used to measure the ability of the receiver to **accept** a given band of frequencies and **reject** all others.

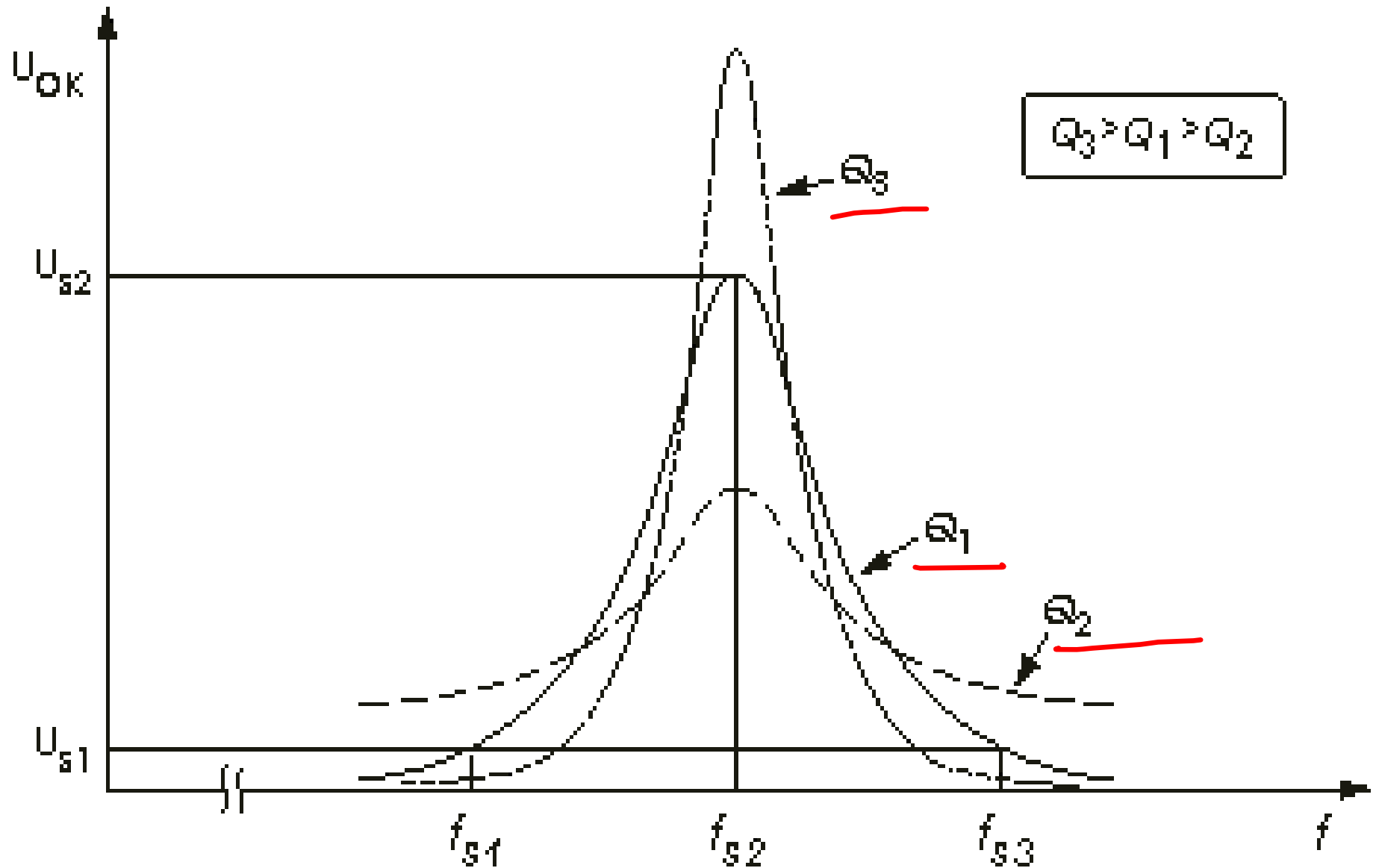
- For commercial AM broadcast bandwidth is 10KHz.

- Hence receiver must limit the bandwidth to 10KHz.

Selectivity

- If not, **more than one channels** will be received and demodulated.
- Selectivity determines the adjacent channel rejection of the receiver.
- A radio receivers must be capable of **separating** the desired channels signal without interference.

Selectivity curve



Sensitivity

- It is the ability of the receiver to **amplify weak** signals.
- The **SNR and the power of the signal** at the output of the audio section are used to determine the quality of a received signal.
- It is also called as **receiver threshold**.
- It also depends on receivers **noise figure**. Reducing noise level will increase sensitivity.

③ Dynamic Range

- It is the **input power range** over which the receiver is **useful**.
- Difference in decibels between the **minimum input level necessary** to identify the signal and the input level that will **overdrive the receiver** and produce distortion.
- **Low dynamic range** can cause a desensitizing of RF amplifiers and results in intermodulation distortion.

④

Fidelity

- Ability of radio receiver to produce, an **exact replica** of the original source information.
- Amplitude, frequency, and phase distortion can **deteriorate the fidelity** of communication system.

⑤ Insertion Loss

- Parameter associated with the frequencies that fall in the pass band of the filter.

- It is ratio of the power transferred to a load with a filter in the circuit to the power transferred to a load without filter.

- Filters are constructed from lossy components and because of that signals are attenuated.

R, L, C

⑥ Image Frequency

- Any frequency other than **selected radio frequency**, if allowed to enter a receiver and mix with the LO, will produce **cross-products equal to IF**.
- Once image frequency is mixed can not be filtered out or suppressed.
- Higher IF should be selected to reduce the image frequency effect.

Image Frequency (Ex-I)

IMAGE FREQUENCY

Case i) $\rightarrow$
1560 kHz
(HELLO EARTH)



$$1560 - 1500 = 60 \text{ kHz}$$

60 kHz
(IF)

Case ii) $\rightarrow$
1440 kHz
(HELLO MARS)



60 kHz
(IF)

Image Frequency (Ex-I)

IMAGE FREQUENCY

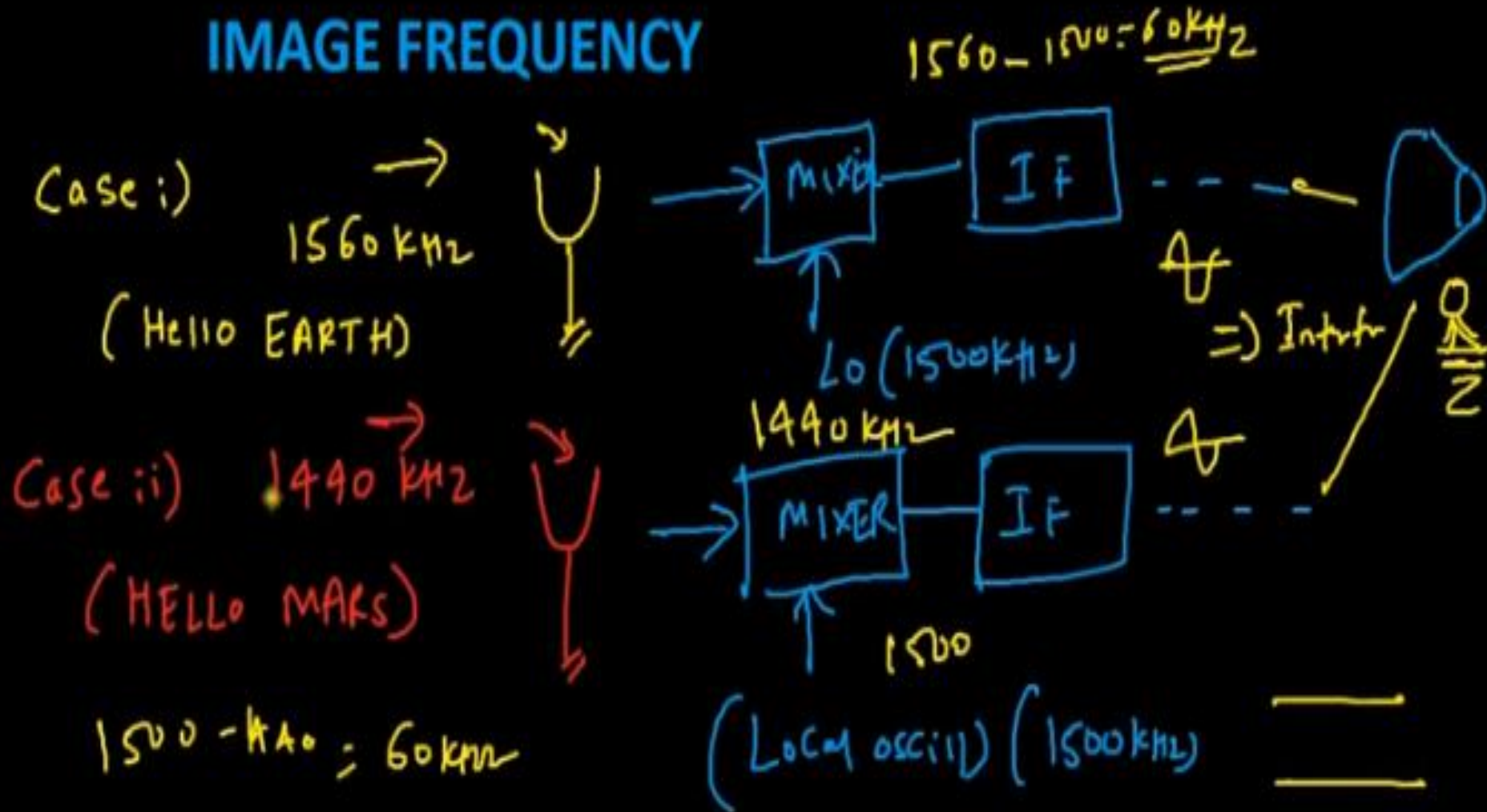


Image Frequency (Ex-I)

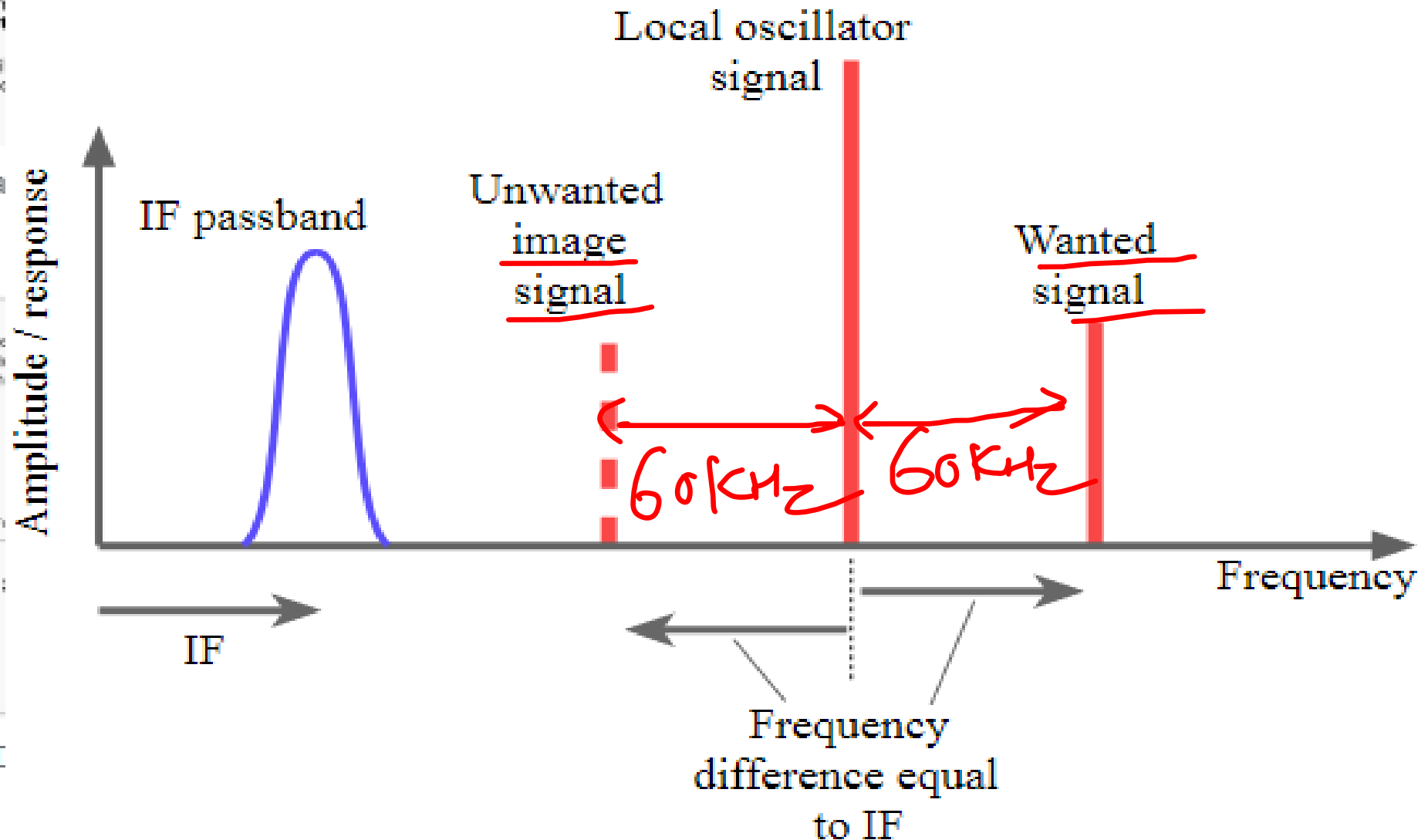


Image Frequency (Ex-I)

$$f_{\text{Image frequency}} = \begin{cases} f + 2I_f & f_{LO} > f \\ f - 2I_f & f_{LO} < f \end{cases}$$

$$f = 1560 \text{ kHz} \quad f_{LO} = 1500 \text{ kHz}$$

$$\begin{aligned} f_{\text{Image}} &= 1560 - 2 \times (1560 - 1500) \\ &= 1560 - 2 \times 60 = 1440 \text{ kHz} \end{aligned}$$

Image Frequency (Ex-2)

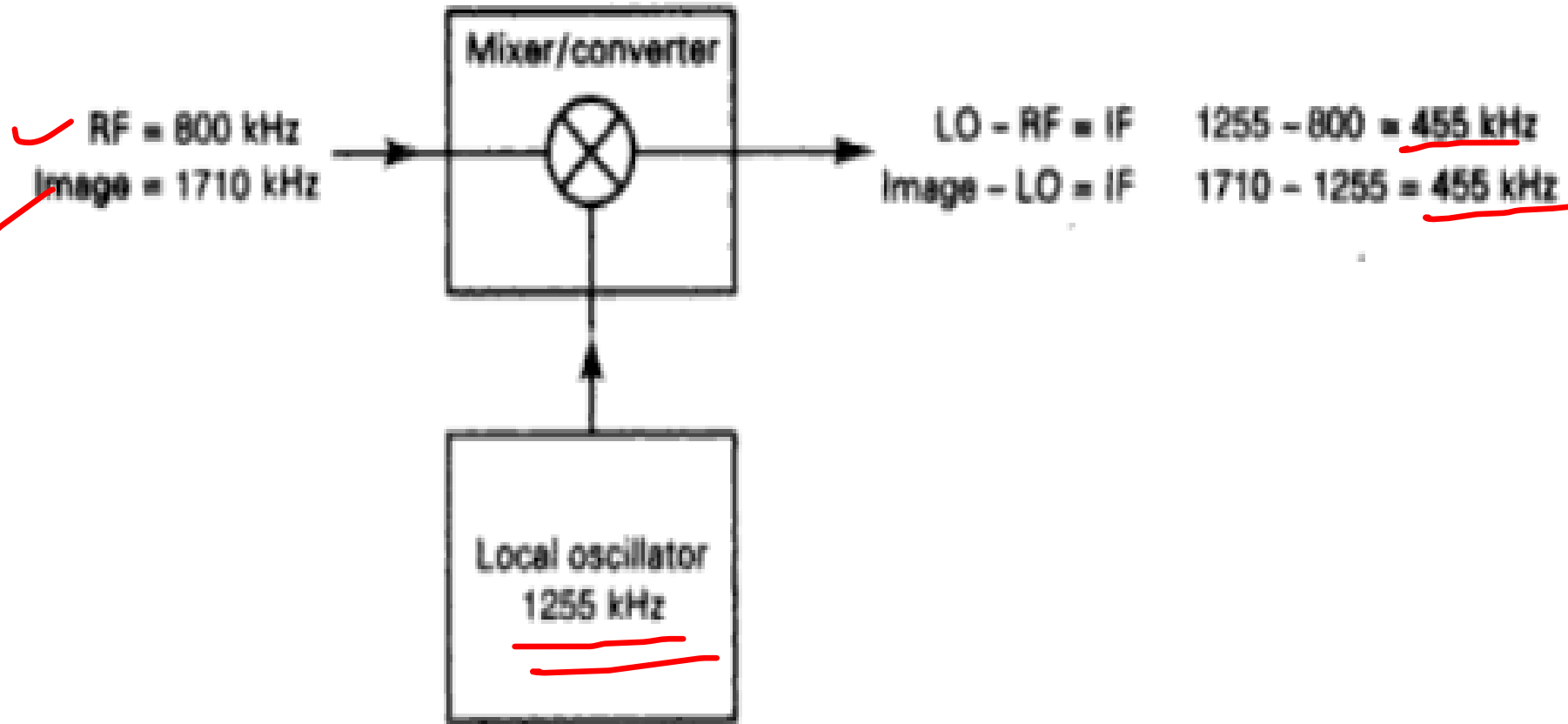


Image Frequency

- For a radio frequency to produce a cross product equal to the IF,
- It must be **displaced from the LO** frequency by a value equal to IF.

Image Frequency

- With high-side injection, the selected RF is below the LO.

$$RF < LO$$



- $f_{image} = f_{RF} + 2f_{IF}$

- With low-side injection, the selected RF is above the LO.

$$RF > LO$$



- $f_{image} = f_{RF} - 2f_{IF}$

Image Frequency Rejection Ratio (IFRR)

- It is numerical measure of the ability of a pre-selector to reject the image frequency.
- For single tuned pre-selector, the ratio of its gain at the desired RF to the gain at the image frequency.

$$IFRR = \sqrt{(1 + Q^2 \rho^2)}$$
$$\text{where, } \rho = \left(\frac{f_{im}}{f_{RF}} \right) - \left(\frac{f_{RF}}{f_{im}} \right)$$

Bandwidth Improvement (BI)

- The noise reduction ratio is achieved by reducing the bandwidth is called **bandwidth improvement (BI)**:

$$BI = \frac{B_{RF}}{B_{IF}}$$

- The corresponding reduction in the noise figure is called **noise figure improvement**:
- **$NF_{improvement} = 10 \log BI$**

Double Spotting

- Wanted station is tuned in at **two spots** on the dial
- Results in **undesired input frequency** to be demodulated by the receiver along with desired incoming signal

Double Spotting

- In this, receiver picks up same station at two nearby frequencies or points on the receiver dial.
- This phenomenon of Double spotting occurs at higher frequencies due to poor front end selectivity of receivers.
- Double spotting is harmful because a weak station may be masked by the reception of a strong station at the same point, on the dial.